New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
  state,
  COUNT(DISTINCT CUSTOMER_ID) as total_customers
FROM
  order_t
JOIN
  customer_t
  USING (CUSTOMER_ID)
GROUP BY
  state
ORDER BY
  total_customers DESC
  LIMIT 10;
```

Output:

Result Grid Filter Rows: Search Export: Fetch rows:

state	total_customers
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26

Result 9

Action Output

	Time	Action	Response
24	15:40:50	SELECT state, COUNT(DISTINCT C...	10 row(s) returned

Solution Query:

```
SELECT
  state,
  COUNT(DISTINCT CUSTOMER_ID) as total_customers
FROM
  order_t
JOIN
  customer_t
  USING (CUSTOMER_ID)
GROUP BY
```

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```
  state
ORDER BY
  total_customers ASC
  LIMIT 10;
```

Output:

Result Grid Filter Rows: Search Export: Fetch rows:

state	total_customers
Wyoming	1
Maine	1
Vermont	1
North Dakota	2
Mississippi	2
New Hampshire	3
Montana	3
New Mexico	5
Arkansas	6
Hawaii	6

Result 10

Action Output

	Time	Action	Response
26	15:50:43	SELECT state, COUNT(DISTINCT C...	10 row(s) returned

Observations and Insights:

- California and Texas have the greatest concentration of customers at 97 each, followed by Florida (86) and New York (69). This indicates demand is tied to more densely populated areas.
- There are several states at a mid-tier concentration (26-35) that show a significant drop off after the top 4 states.
- The states with the least customers are also the lower-population states Wyoming, Maine and Vermont (1 each), indicating demand is not evenly distributed throughout the US.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
        VEHICLE_MAKER,
        COUNT(DISTINCT CUSTOMER_ID) as total_customers
FROM
        product_t
JOIN
        order_t
        USING (PRODUCT_ID)
GROUP BY
        VEHICLE_MAKER
ORDER BY
        total_customers DESC
LIMIT 10;
```

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Output:

Result Grid Filter Rows: Search Export: Fetch rows:

VEHICLE_MAKER	total_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50
Mercedes-Benz	45
Mazda	43
Mitsubishi	41
Buick	40
GMC	37

Result 13 Result 14

Action Output

	Time	Action	Response
35	16:02:42	SELECT VEHICLE_MAKER, COUNT(DI...	10 row(s) returned

Observations and Insights:

- The top vehicle makers are Chevrolet (83), Ford (63) and Toyota (52).
- Pontiac and Dodge are tied at 50.
- The following vehicle makers steadily decrease in the amount of customers, indicating demand is strongest within the dominant vehicle makers.

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Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
        STATE,
        VEHICLE_MAKER,
   total_customers
FROM
        (SELECT
                c.STATE,
                 p.VEHICLE_MAKER,
                COUNT(DISTINCT o.CUSTOMER_ID) AS total_customers,
                RANK () OVER (
                PARTITION BY c.STATE
                ORDER BY COUNT(DISTINCT o.CUSTOMER_ID) DESC) AS rnk
FROM order_t o
JOIN customer_t c
        ON o.CUSTOMER_ID = c.CUSTOMER_ID
JOIN product_t p
        ON p.PRODUCT_ID = o.PRODUCT_ID
GROUP BY
        c.STATE,
        p.VEHICLE_MAKER) ranked_data
WHERE rnk = 1
ORDER BY STATE;
```

Output:

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Result Grid Filter Rows: Search Export:

STATE	VEHICLE_MAKER	total_customers
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Chevrolet	1
Arkansas	GMC	1
Arkansas	Mitsubishi	1
Arkansas	Pontiac	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
California	Nissan	6
California	Audi	6
California	Chevrolet	6
California	Dodge	6
California	Ford	6
Colorado	Chevrolet	5
Connec…	Maserati	2
Connec…	Chevrolet	2
Connec…	Volvo	2
Connec…	Mercury	2
Delaware	Mitsubishi	2
District…	Chevrolet	4
Florida	Toyota	7
Georgia	Toyota	3
Hawaii	Ford	1
Hawaii	GMC	1
Hawaii	Nissan	1
Hawaii	Pontiac	1
Hawaii	Toyota	1
Hawaii	Cadillac	1
Idaho	Dodge	2
Illinois	GMC	3
Illinois	Ford	3
Illinois	Chevrolet	3
Indiana	Mazda	4
Iowa	Chevrolet	1
Iowa	Chrysler	1
Iowa	Dodge	1
Iowa	Ford	1
Iowa	Hyundai	1
Iowa	Isuzu	1
Iowa	Jeep	1

Result 15 | Result 16 | Result 17

Action Output

	Time	Action	Response
40	16:57:11	SELECT STATE, VEHICLE_MAKER, total_custome…	143 row(s) returned

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STATE	VEHICLE_MAKER	total_customers
Iowa	Jeep	1
Iowa	Mazda	1
Iowa	Pontiac	1
Iowa	Porsche	1
Iowa	Subaru	1
Kansas	GMC	1
Kansas	Dodge	1
Kansas	Ford	1
Kansas	Saab	1
Kansas	Honda	1
Kansas	Lexus	1
Kansas	Maserati	1
Kansas	Mazda	1
Kansas	Mercedes-Benz	1
Kansas	Nissan	1
Kansas	Buick	1
Kansas	Suzuki	1
Kansas	Volkswagen	1
Kentucky	Acura	1
Kentucky	Audi	1
Kentucky	Mercedes-Benz	1
Kentucky	Mercury	1
Kentucky	Nissan	1
Kentucky	Pontiac	1
Kentucky	Ram	1
Kentucky	Volvo	1
Louisiana	Kia	2
Louisiana	BMW	2
Louisiana	Nissan	2
Louisiana	Pontiac	2
Louisiana	Ford	2
Maine	Mercedes-Benz	1
Maryland	Ford	5
Massac…	Dodge	2
Massac…	Chevrolet	2
Michigan	Ford	3
Minnesota	GMC	3
Mississi…	Dodge	1
Mississi…	Toyota	1
Missouri	Chevrolet	4
Montana	Chevrolet	1
Montana	Dodge	1

Result 15 | Result 16 | Result 17

Action Output

	Time	Action	Response
40	16:57:11	SELECT STATE, VEHICLE_MAKER, total_custome...	143 row(s) returned

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Result Grid Filter Rows: Search Export:

STATE	VEHICLE_MAKER	total_customers
Montana	Dodge	1
Montana	Mitsubishi	1
Nebraska	Mercedes-Benz	1
Nebraska	Chevrolet	1
Nebraska	Cadillac	1
Nebraska	Nissan	1
Nebraska	Pontiac	1
Nebraska	Toyota	1
Nebraska	Volkswagen	1
Nevada	Pontiac	3
New Ha…	Chrysler	1
New Ha…	Lexus	1
New Ha…	Lincoln	1
New Je…	Hyundai	2
New Je…	Mercedes-Benz	2
New M…	Dodge	2
New York	Toyota	5
New York	Pontiac	5
North C…	Volvo	3
North D…	Ford	1
North D…	Hyundai	1
Ohio	Chevrolet	6
Oklahoma	Mazda	2
Oklahoma	Toyota	2
Oklahoma	Ferrari	2
Oregon	Toyota	2
Pennsyl…	Toyota	3
South C…	Acura	1
South C…	BMW	1
South C…	Buick	1
South C…	Dodge	1
South C…	Isuzu	1
South C…	Jaguar	1
South C…	Kia	1
South C…	Mazda	1
South C…	Mitsubishi	1
Tennes…	Mazda	3
Texas	Chevrolet	9
Utah	Lincoln	1
Utah	Chevrolet	1
Utah	Dodge	1
Utah	Isuzu	1

Result 15 | Result 16 | Result 17

ction Output

	Time	Action
40	16:57:11	SELECT STATE, VEHICLE_MAKER, total_custome… 1

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Utah	Dodge	1
Utah	Isuzu	1
Utah	Buick	1
Utah	Maybach	1
Utah	Oldsmobile	1
Utah	Pontiac	1
Utah	Subaru	1
Utah	Volkswagen	1
Vermont	Mazda	1
Virginia	Ford	5
Washin…	Chevrolet	5
West Vi…	Mercedes-Benz	2
Wisconsin	Acura	1
Wisconsin	Cadillac	1
Wisconsin	Chevrolet	1
Wisconsin	Dodge	1
Wisconsin	Honda	1
Wisconsin	Mazda	1
Wisconsin	Nissan	1
Wisconsin	Pontiac	1
Wyoming	Buick	1

Result 15 | Result 16 | Result 17

ction Output

	Time	Action
40	16:57:11	SELECT STATE, VEHICLE_MAKER, total_custome...

Observations and Insights:

- The most popular vehicle makers vary across states showing regional differences in customer preferences.
- The top vehicle makers Chevrolet, Ford and Toyota have a consistent presence throughout the states and regions.
- Some states show a clear preference for one vehicle maker while others favor multiple vehicle makers equally.

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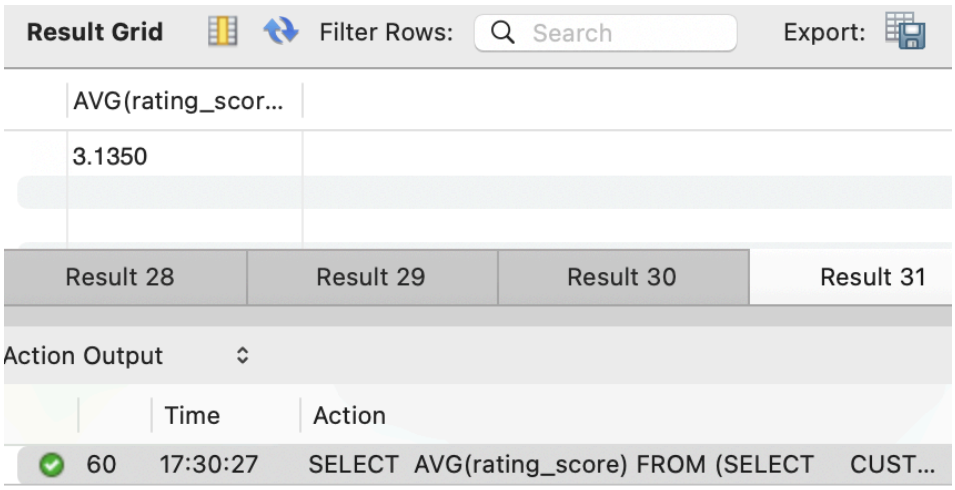
Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
        AVG(rating_score)
FROM
(SELECT
   CUSTOMER_FEEDBACK,
   CASE
      WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
      WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 2
      WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 3
      WHEN CUSTOMER_FEEDBACK = 'Good' THEN 4
      WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 5
   END AS rating_score
FROM order_t) AS ratings;
```

Output:



Solution Query:

```
SELECT
        QUARTER_NUMBER,
        AVG(rating_score) AS avg_rating
FROM
(SELECT
        QUARTER_NUMBER,
        CUSTOMER_FEEDBACK,
```

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```
        CASE
                WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
                WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 2
                WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 3
                WHEN CUSTOMER_FEEDBACK = 'Good' THEN 4
                WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 5
        END AS rating_score
FROM order_t) AS ratings
GROUP BY
QUARTER_NUMBER
ORDER BY
QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	avg_rating
1	3.5548
2	3.3550
3	2.9563
4	2.3970

Result 44 | Result 45 | Result 46 | Result 47

Action Output

	Time	Action	Response
85	17:38:40	SELECT QUARTER_NUMBER, AVG(rating_score) AS...	4 row(s) returned

Observations and Insights:

- The average customer feedback score is 3.14 indicating a moderate amount of satisfaction better than 'Okay'.
- The average customer rating begins in Q1 at 3.55, showing a moderately positive experience for customers and ends at 2.40 indicating a somewhat negative experience using New Wheels.
- The sharpest decrease in customer satisfaction occurs between Q3 and Q4 and is possibly related to the increased shipping time and operational delays.

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Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
   QUARTER_NUMBER,
   SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN 1 ELSE 0 END)
      / COUNT(*) * 100 AS very_bad_pct,
   SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 1 ELSE 0 END)
      / COUNT(*) * 100 AS bad_pct,
   SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 1 ELSE 0 END)
      / COUNT(*) * 100 AS okay_pct,
   SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Good' THEN 1 ELSE 0 END)
      / COUNT(*) * 100 AS good_pct,
   SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 1 ELSE 0 END)
      / COUNT(*) * 100 AS very_good_pct
FROM order_t
GROUP BY
   QUARTER_NUMBER
ORDER BY
   QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_p...
1	10.9700	11.2900	19.0300	28.7100	30.0000
2	14.8900	14.1200	20.2300	22.1400	28.6300
3	17.9000	22.7100	21.8300	20.9600	16.5900
4	30.6500	29.1500	20.1000	10.0500	10.0500

Result 48 | Result 49 | Result 50 | Result 51 | Result 52

Action Output

	Time	Action
91	19:58:10	SELECT QUARTER_NUMBER, SUM(CASE WHEN CUSTOMER_FEEDBACK =

Observations and Insights:

- During Q1, customer feedback dominates with close to 60% having 'Good' or 'Very Good' experience. Negative feedback during this time is minimal, close to 22%.
- Q2 shows positive feedback declining as negative feedback increases somewhat.
- In Q3, negative feedback begins to dominate at close to 41%, while positive feedback decreases to close to 37%.
- By Q4, negative feedback (~60%) significantly outweighs positive feedback at about 20%.

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- This trend suggests that customer experience deteriorated over time, likely due to longer shipping times and operational issues.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
        QUARTER_NUMBER,
   COUNT(ORDER_ID) AS total_orders
FROM
   order_t
GROUP BY
        QUARTER_NUMBER
ORDER BY
        QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	total_orders
1	310
2	262
3	229
4	199

Result 53 | Result 54 | Result 55 | Result 56

Action Output

	Time	Action
98	20:05:21	SELECT QUARTER_NUMBER, COUNT(ORDER_ID)

Observations and Insights:

- The total amount of orders began strong at 310 within Q1 and decreased consistently with each quarter indicating a consistent loss of customer engagement.
- By Q4, orders decreased by close to 36%, likely due to shipping delays and loss of customer satisfaction.

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Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
        QUARTER_NUMBER,
        NET_REVENUE,
         PREV_NET_REVENUE,
CASE
        WHEN PREV_NET_REVENUE IS NULL OR PREV_NET_REVENUE = 0 THEN NULL
        ELSE (NET_REVENUE - PREV_NET_REVENUE)*100/ PREV_NET_REVENUE
END AS QUARTER_PCT_CHANGE
FROM (
        SELECT
                QUARTER_NUMBER,
                NET_REVENUE,
                LAG(NET_REVENUE) OVER (ORDER BY QUARTER_NUMBER) AS PREV_NET_REVENUE
        FROM (
                SELECT
                        QUARTER_NUMBER,
                        SUM(QUANTITY * VEHICLE_PRICE * (1 - DISCOUNT / 100)) AS NET_REVENUE
                FROM order_t
                GROUP BY QUARTER_NUMBER) QREV) QREV_LAG
ORDER BY QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	NET_REVENUE	PREV_NET_REVENUE	QUARTER_PCT_CHANGE
1	39421580.15929600	NULL	NULL
2	32715830.33996200	39421580.15929600	-17.010352685603
3	29229896.19364900	32715830.33996200	-10.655190805458
4	23346779.63060600	29229896.19364900	-20.127052535757

Result 84 | Result 85 | Result 86 | Result 87 | Result 88 | Resu

Action Output

	Time	Action	Response
138	20:53:03	SELECT QUARTER_NUMBER, NET_REVENUE,...	4 row(s) returned

Observations and Insights:

- The net revenue is strong in Q1 at $39.4M and consistently declines each quarter hitting $23.3M by Q4.
- In Q2, New Wheels lost 17% net revenue, followed by 10.7% in Q3 and 20% in Q4, indicating worsening performance throughout the year.

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- The loss of net revenue by Q4 is consistent with delivery delays, decreased customer satisfaction and fewer orders placed.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
        o.QUARTER_NUMBER,
        SUM(o.QUANTITY * o.VEHICLE_PRICE * (1 - o.DISCOUNT / 100)) AS net_revenue,
        COUNT(o.ORDER_ID) AS total_orders
FROM order_t o
GROUP BY o.QUARTER_NUMBER
ORDER BY o.QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	net_revenue	total_orders
1	39421580.15929600	310
2	32715830.33996200	262
3	29229896.19364900	229
4	23346779.63060600	199

Result 169 | Result 170 | Result 171 | Result 172

Action Output

	Time	Action
247	18:34:04	SELECT o.QUARTER_NUMBER, SUM(o.QUANTITY *...

Observations and Insights:

- The first quarter begins with a net revenue of $39.4M and ends at $23.3M, going from 310 total orders to 199 total orders within that same window of time.
- The decrease in orders along with net revenue suggests customer demand is a major driver of financial performance.
- The continuous decrease in customer orders and net revenue are congruent with decreasing customer satisfaction and challenges with order fulfillment in shipping delays.

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Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
        AVG(o.DISCOUNT) AS avg_discount_percent,
         c.CREDIT_CARD_TYPE
FROM
        order_t o
JOIN
        customer_t c
        USING (CUSTOMER_ID)
GROUP BY c.CREDIT_CARD_TYPE
ORDER BY avg_discount_percent DESC;
```

Output:

Result Grid Filter Rows: Search

avg_discount_perc...	CREDIT_CARD_TYPE
0.643846	laser
0.629500	mastercard
0.624219	maestro
0.623469	visa-electron
0.622174	china-unionpay
0.620625	instapayment
0.616327	americanexpress
0.614615	diners-club-us-ca
0.614490	diners-club-carte-blanche
0.610233	switch
0.609545	bankcard
0.607382	jcb
0.600833	visa
0.599792	diners-club-enroute
0.585000	solo
0.584000	diners-club-international

Result 33 | Result 34 | Result

Action Output

	Time	A	Response
50	20:38:28	SI	16 row(s) returned

Observations and Insights:

- The average discount percent between credit card types ranges from %64 -58, staying within a fairly narrow range.
- Customers using Laser and Mastercard reap the largest discounts at %64-63, while customers paying with Solo and Diner's Club receive the smallest discounts at %58.

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- In order to increase sales and customer engagement, New Wheels could offer targeted promotions for cards with the least amount of discount.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
        o.QUARTER_NUMBER,
        AVG(DATEDIFF(o.SHIP_DATE,o.ORDER_DATE)) AS avg_ship_time
FROM
        order_t o
WHERE o.SHIP_DATE IS NOT NULL AND o.ORDER_DATE IS NOT NULL
GROUP BY o.QUARTER_NUMBER
ORDER BY o.QUARTER_NUMBER;
```

Output:

Result Grid Filter Rows: Search Export:

QUARTER_NUMBER	avg_ship_time
1	57.1677
2	71.1107
3	117.7555
4	174.0955

Result 260 | Result 261 | Result 262 | Result 263 | Resu

Action Output

	Time	Action	Response
351	19:02:01	SELECT o.QUARTER_NUMBER,...	4 row(s) returned

Observations and Insights:

- The average shipping time steadily increases throughout the year, with the sharpest increases occurring after Q2.
- The average order-to-ship time for the first quarter is about ~57 days. By the second quarter, it increased by ~14 days. The third quarter shows a sharper increase of about ~46 days, with the last quarter showing the most significant increase at about ~56 days.
- This shows a consistent decrease in shipping performance, possibly indicating operational bottlenecks.

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Business Metrics Overview

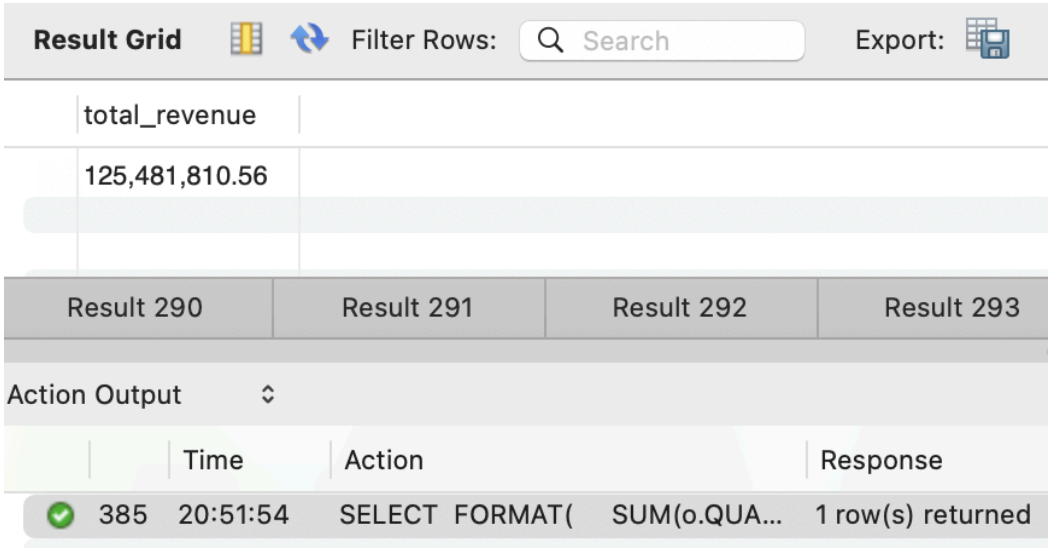
Total Revenue	Total Orders	Total Customers	Average Rating
$125,481,810.56	1000	994	3.135
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
$29,435,200.08	229	98	44.1%

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Solution Query:

```
SELECT
        FORMAT(
        SUM(o.QUANTITY * o.VEHICLE_PRICE - (1 - o.DISCOUNT / 100)),2)
         AS total_revenue
FROM
        order_t o;
```

Output:



Solution Query:

```
SELECT
        COUNT(DISTINCT o.ORDER_ID)
FROM
        order_t o;
```

Output:

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Result Grid Filter Rows: Search Export:

COUNT(DISTINCT o.ORDER_ID)
1000

Result 312 | Result 313 | Result 314 | Result 315

Action Output

	Time	Action	Response
411	20:58:40	SELECT COUNT(DISTINCT o.ORD...	1 row(s) returned

Solution Query:

```
SELECT
        COUNT(DISTINCT o.CUSTOMER_ID) AS total_customers
FROM
        order_t o;
```

Output:

Result Grid Filter Rows: Search

total_customers
994

Result 324 | Result 325 | Result 326

Action Output

	Time	Action
425	21:18:57	SELECT COUNT(DISTINCT o.CUS...

Solution Query (for last Quarter Revenue):

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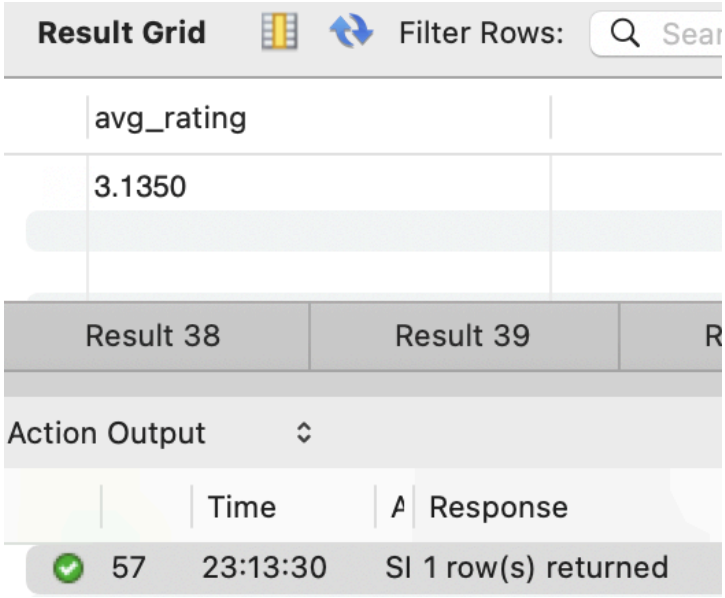
```
SELECT
  AVG(rating_score) AS avg_rating
FROM(
  SELECT
    CASE
      WHEN CUSTOMER_FEEDBACK = 'Very Bad'  THEN 1
      WHEN CUSTOMER_FEEDBACK = 'Bad'       THEN 2
      WHEN CUSTOMER_FEEDBACK = 'Okay'      THEN 3
      WHEN CUSTOMER_FEEDBACK = 'Good'      THEN 4
      WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 5
    END AS rating_score
  FROM order_t) AS rating;
```

Output:



Solution Query (for last Quarter Orders):

```
SELECT
  QUARTER_NUMBER,
  COUNT(DISTINCT ORDER_ID)
FROM order_t
WHERE QUARTER_NUMBER = 3;
```

Output:

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Result Grid Filter Rows: Search

QUARTER_NUMBER	COUNT(DISTINCT ORDER_ID)
3	229

Result 44 | Result 45 | Result 46

Action Output

	Time	Response
65	23:14:19	SI 1 row(s) returned

Query Solution (average shipping time):

```
SELECT
  AVG(avg_ship) AS avg_shipping_time
FROM(
  SELECT DATEDIFF(SHIP_DATE, ORDER_DATE) AS avg_ship
  FROM order_t) AS shiptime;
```

Output:

Result Grid Filter Rows: Search

avg_shipping_time
97.9640

Result 4 | Result 5 | Result 6

Action Output

	Time	Response
12	19:36:14	SI 1 row(s) returned

Solution Query (for % Good Feedback):

```
SELECT
  SUM(CASE
      WHEN CUSTOMER_FEEDBACK IN ('Good', 'Very Good') THEN 1
      ELSE 0
```

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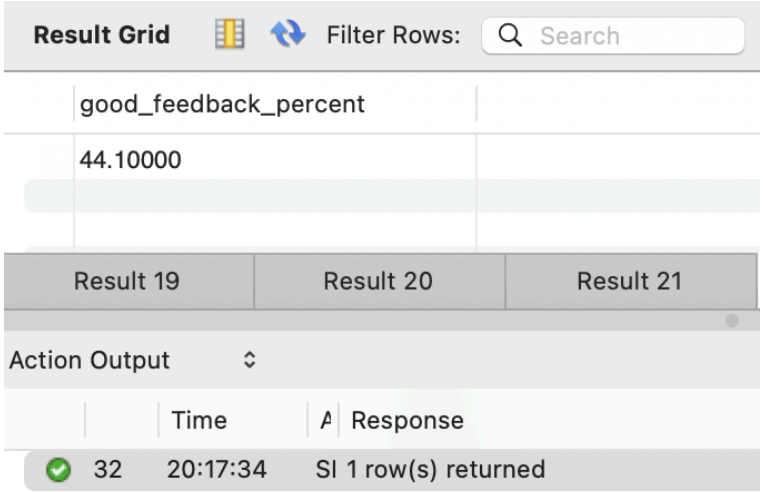
```
    END) * 100.0
  / COUNT(*) AS good_feedback_percent
FROM order_t;
```

Output:



Business Recommendations

- In order to reverse declining sales and customer engagement, New Wheels should focus on improving order fulfillment and delivery efficiency. This could mean monitoring fulfillment metrics and logistics.
- New Wheels should also prioritize customer experience by investing in customer support and proactive communication to rebuild trust. They could also introduce order tracking to further optimize customer experience.
- Targeted promotions and loyalty programs could help increase regular customer engagement and drive sales, stabilizing revenue.

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